

CHRISTOPHER AVINA

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SUMMARY

Results driven and adaptable engineer with hands-on experience in software development, data engineering, and machine learning. I am proficient in Python, C++, Java, and skilled in ETL pipelines, cloud computing, game development, and big data technologies. With a proven track record of solving complex problems and adapting to new technologies, I am eager to apply my technical skillset and passion for data-driven solutions to contribute to innovative projects.

TECHNICAL SKILLS

PROGRAMMING LANGUAGES:

Python, SQL, C, C++, Java, Javascript

FRAMEWORKS/LIBRARIES:

PySpark, TensorFlow, OpenCV, Flask, Django, SKLearn, MERN Stack, NumPy, Matplotlib

DATA ENGINEERING & CLOUD:

AWS (S3, EC2, Lambda), Databricks, Snowflake, Apache Spark, Airflow, PostgreSQL, MySQL, MongoDB

EMBEDDED & OTHER:

ARM/RISC-V (STM32, ATmega32x, ESP32), Systems Communication (I2C, SPI, UART, BLE), PCB Design (KiCad), Circuit Design & Analysis

EXPERIENCE

Software Engineer | Starcraft 2 Reinforcement Learning AI (Freelance) *(Jan 2025-Present)*

- Developed a Zerg AI using reinforcement learning (Q-learning table) for strategic decision-making.
- Leveraged DeepMind's PySC2 for feature extraction and game environment analysis.
- Optimized reward functions for improved drone production, tech progression, and combat engagements.
- Implementing transformer-based language model to generate explanations of AI decision-making.

Data Engineer | NASA ELT Pipeline, (Freelance) *(August 2024-Sept 2024)*

- Designed and implemented a medallion architecture data pipeline to extract and store NASA API data.
- Utilized AWS S3 for structured storage and PySpark for ETL transformations.
- Developed interactive dashboards using Databricks, NumPy, and Matplotlib.
- Automated the ETL process with Airflow Workflows, reducing manual intervention and improving efficiency

Embedded Software Engineer | Rekovar *(April 2023 - June 2023)*

- Led a comprehensive overhaul of project structure and build systems, increasing engineer productivity by
- Developed and debugged neonatal medical device software, collaborating with offshore engineers.
- Increased unit test coverage from 0% to 100% using Ceedling, Unity, and CMock.
- Created documentation and SOPs for FDA compliance via Greenlight Guru.
- Left due to cancer treatments; now cancer-free.

PROJECTS

Software Engineer | Breath of Fire 3: Teepo's Revenge (Open Source Contributor) *(Aug 2023- March 2024)*

- Optimized battle engine by improving data access through data structure and algorithm complexities.
- Tracked and fixed bugs across the game as necessary.
- Project is now defunct.

Lead Engineer | BabyWatch (UCI Capstone Project) *(Sep 2022- March 2023)*

- Led a team to develop an edge-computing media server on a Jetson Nano for real-time infant monitoring.
- Trained machine learning models (SKLearn, OpenCV) to detect unsafe baby positions (70% success rate).
- Deployed AES-128 encryption to secure video footage from unauthorized access.
- Built a Flask web application for real-time video streaming and data visualization.

EDUCATION

UNIVERSITY OF CALIFORNIA, IRVINE

B.S. Computer Science & Engineering | Spring 2023

RELEVANT LINKS

- NASA Data Pipeline: github.com/Cavina/NasaRoverProject
- ZergBot - Reinforcement Learning AI: github.com/Cavina/ZergBot
- BabyWatch - Machine Learning: github.com/Cavina/Baby_Machine_Learning